

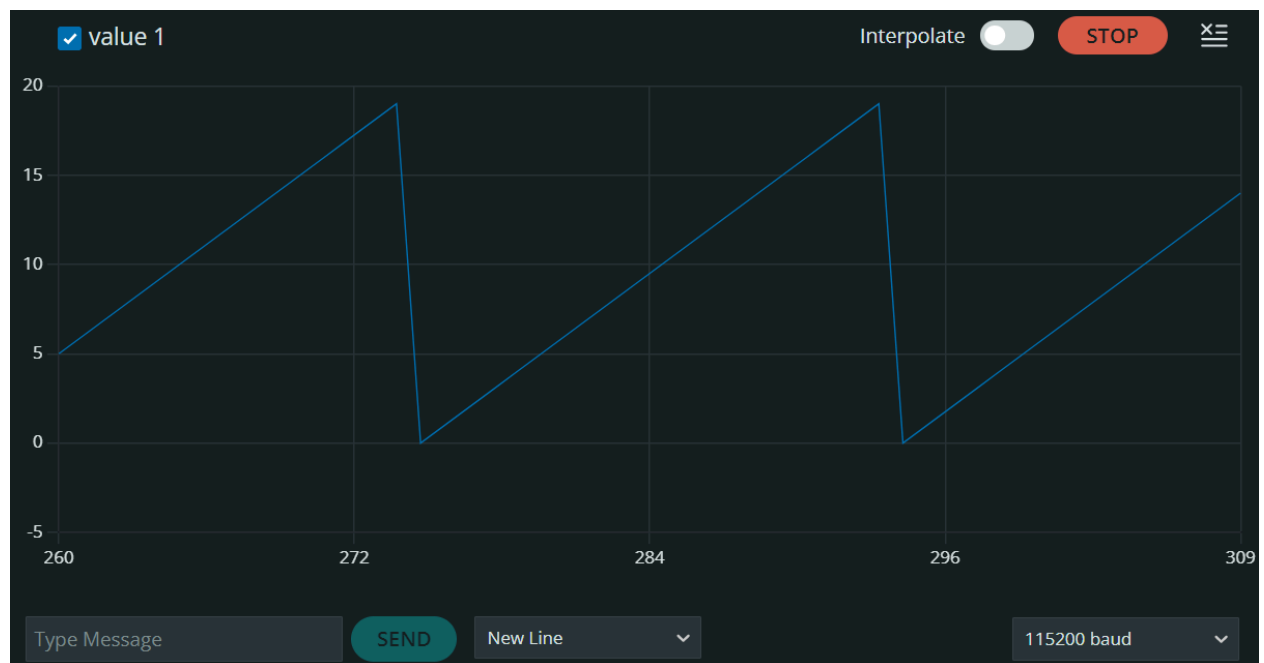
EE446-TinyML Lab 2

Author: Sparsh Dadhich ID: 2239751

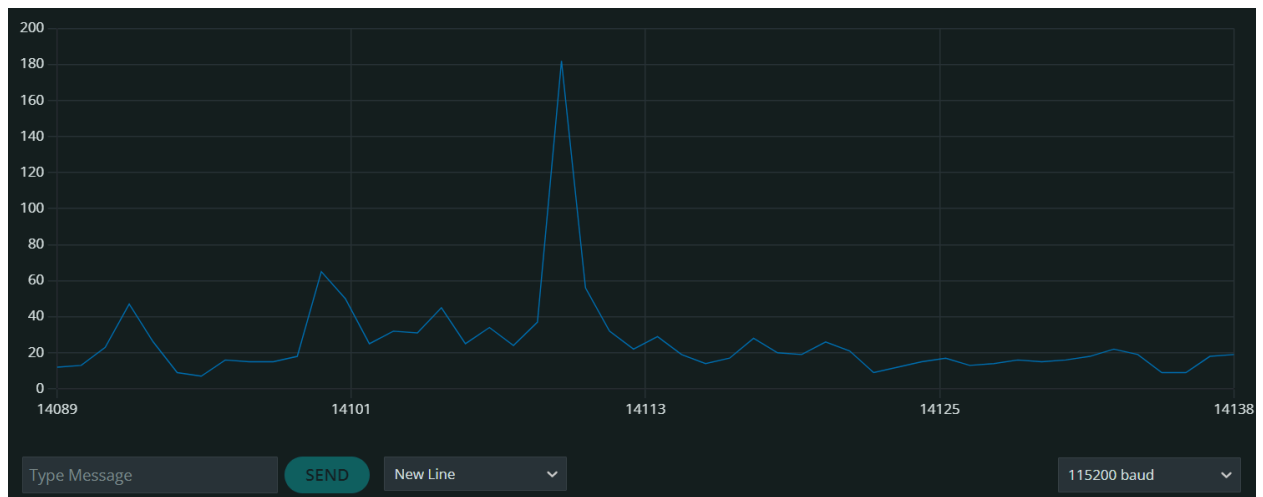
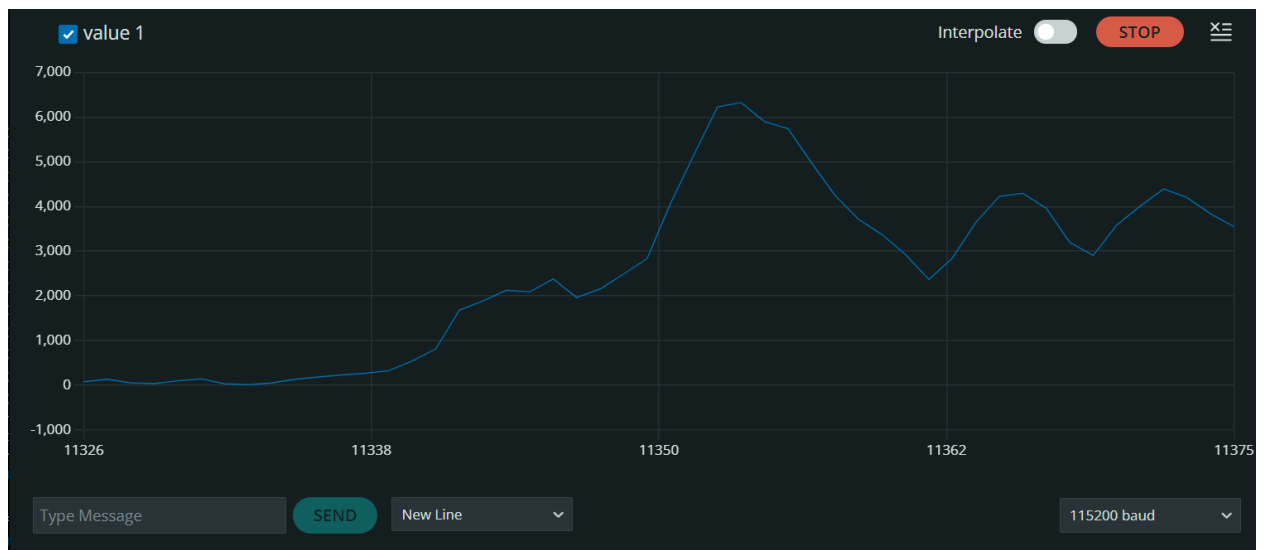
Github: <https://github.com/sdadhich04/EE-446-TinyML->

Task 4 :

```
Serial test started  
Count: 0  
Count: 1  
Count: 2  
Count: 3  
Count: 4  
Count: 5  
Count: 6  
Count: 7
```

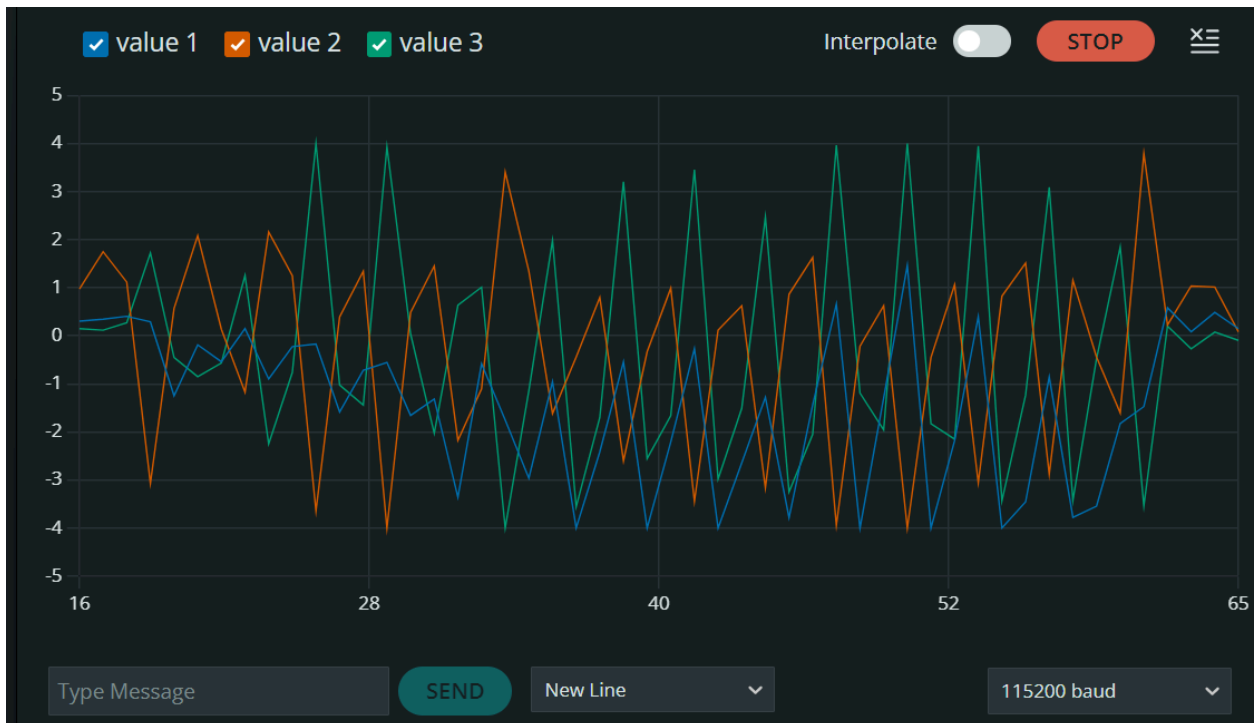


Task 5 : The cleanest change was created when I softly clapped near the MCU. It is the bottom most plot which shows a clear distinct sign of the event happening.



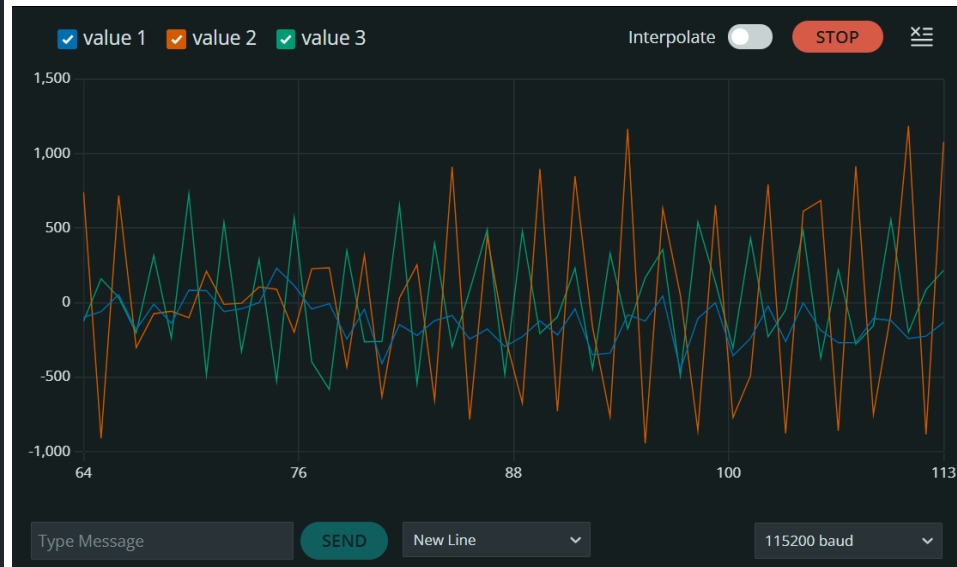
Task 6A : It is hard to say which orientation produced the biggest change but if I were to say which was the cleanest one it would be when i flipped the board upside down

-0.008,0.047,1.007	0.981,-0.299,-0.057	-1.001,0.009,0.110
-0.007,0.047,1.009	0.953,-0.232,0.077	-1.006,-0.003,0.075
-0.007,0.045,1.004	0.897,-0.124,0.061	-1.006,0.020,0.071
-0.007,0.045,1.008	0.907,-0.180,0.060	-1.005,0.010,0.059
-0.003,0.045,1.005	1.015,-0.183,0.233	-1.009,0.004,0.070
-0.006,0.046,1.008	0.952,-0.200,0.057	-1.009,0.006,0.042
-0.004,0.047,1.009	0.919,-0.159,0.099	-1.005,0.009,0.065
-0.009,0.046,1.007	0.945,-0.134,0.156	-1.004,0.006,0.070
-0.005,0.050,1.007	0.966,-0.206,0.157	-1.004,0.005,0.055
-0.007,0.048,1.009	0.953,-0.205,0.138	-1.006,0.016,0.060
-0.007,0.047,1.007	0.941,-0.185,0.138	-1.002,0.016,0.059
-0.005,0.046,1.007	0.957,-0.191,0.162	-1.006,0.003,0.041
-0.006,0.048,1.006	0.963,-0.219,0.137	-1.007,-0.000,0.044
-0.009,0.046,1.007	0.947,-0.179,0.147	-1.004,0.009,0.056
-0.003,0.049,1.007	0.960,-0.190,0.158	-1.003,0.013,0.062
-0.005,0.046,1.005	0.950,-0.150,0.156	-1.002,0.025,0.077
-0.007,0.045,1.006	0.960,-0.200,0.135	-1.006,0.017,0.069
-0.005,0.044,1.007	0.956,-0.141,0.178	-1.006,0.007,0.061
-0.006,0.047,1.007	0.961,-0.180,0.137	-1.006,0.013,0.062
-0.007,0.048,1.009	0.954,-0.167,0.148	-1.002,0.022,0.080
-0.003,0.047,1.009	0.967,-0.199,0.137	-1.004,0.019,0.073
-0.015,0.047,1.011	0.955,-0.189,0.137	-1.002,0.017,0.063
-0.009,0.045,1.007	0.968,-0.204,0.148	-1.008,0.018,0.043
-0.006,0.046,1.003	0.962,-0.188,0.154	
-0.006,0.047,1.011	0.962,-0.211,0.131	
-0.008,0.047,1.007		
-0.008,0.048,1.008		



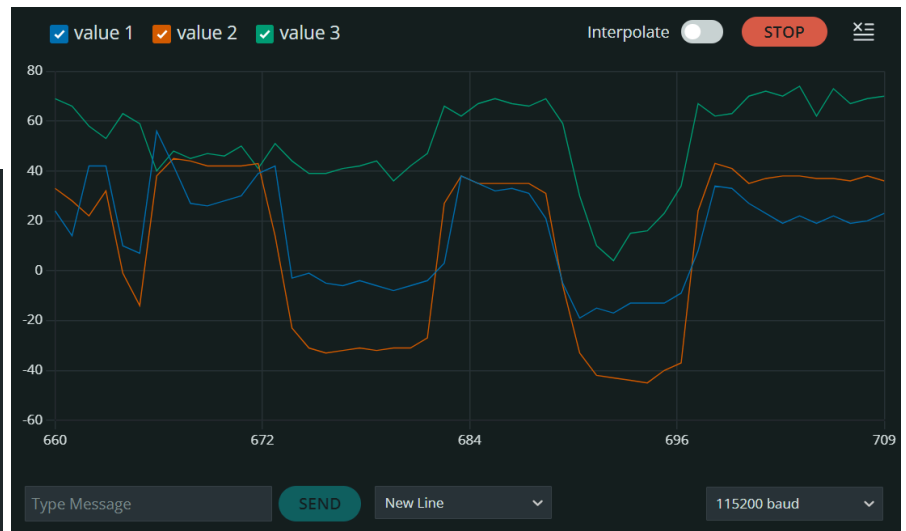
Task 6B: The cleanest change was when I rotated the board along its vertical axis.

```
1.160,0.061,0.000
0.793,0.000,0.000
-0.061,0.427,-0.366
0.793,0.000,0.061
2.625,-0.061,0.488
3.296,-0.366,0.854
1.953,0.183,0.366
2.502,-0.305,0.549
0.854,-0.122,-0.122
0.793,0.122,0.244
-0.549,0.061,-0.488
1.587,-0.183,0.305
0.671,-0.122,-0.244
-1.160,0.183,-0.732
0.122,0.366,-0.244
0.732,0.000,-0.061
2.319,0.000,0.488
0.854,0.000,0.000
-1.099,0.366,-0.610
-2.258,0.610,-0.916
-0.427,-0.061,-0.671
-0.183,0.366,0.061
2.930,-0.366,0.671
0.732,0.183,0.061
-0.854,0.427,-0.305
-0.305,-0.671,-1.099
1.953,-0.366,0.244
1.831,-0.061,0.793
1.343,0.183,0.366
0.305,0.244,0.061
-0.061,0
```



Task 6C : The cleanest change in the magnetometer readings was when I bought it close to my laptop.

```
45.000,-23.000,16.000
44.000,-26.000,17.000
45.000,-23.000,16.000
42.000,-23.000,22.000
34.000,-24.000,23.000
32.000,-21.000,23.000
32.000,-21.000,29.000
33.000,-22.000,25.000
34.000,-20.000,24.000
35.000,-21.000,29.000
34.000,-22.000,24.000
```



Task 7 : The cleanest change was noticed when I blew hot air onto the board which produced a huge spike in the Humidity levels.

Temperature (C): 25.64	Humidity (%): 26.13
Temperature (C): 25.64	Humidity (%): 26.13
Temperature (C): 25.62	Humidity (%): 26.13
Temperature (C): 25.62	Humidity (%): 26.15
Temperature (C): 25.61	Humidity (%): 26.15
Temperature (C): 25.61	Humidity (%): 26.16
Temperature (C): 25.61	Humidity (%): 26.19
Temperature (C): 25.60	Humidity (%): 26.21
Temperature (C): 25.56	Humidity (%): 26.23
Temperature (C): 25.56	Humidity (%): 26.21
Temperature (C): 25.55	Humidity (%): 26.19
Temperature (C): 25.53	Humidity (%): 26.13
Temperature (C): 25.53	Humidity (%): 26.10
Temperature (C): 25.53	Humidity (%): 26.06

Temperature (C): 27.53	Humidity (%): 43.05
Temperature (C): 26.44	Humidity (%): 58.82
Temperature (C): 27.47	Humidity (%): 67.45
Temperature (C): 28.33	Humidity (%): 73.35
Temperature (C): 27.85	Humidity (%): 75.90
Temperature (C): 28.84	Humidity (%): 79.37
Temperature (C): 28.64	Humidity (%): 80.62
Temperature (C): 28.68	Humidity (%): 80.91
Temperature (C): 28.87	Humidity (%): 81.06
Temperature (C): 28.95	Humidity (%): 82.09

Task 8 : The cleanest change in the pressure value was when I measured it at the top floor of Accolade apartments where the pressure value decreased.

Pressure (kPa): 101.662	Temperature (C): 26.63
Pressure (kPa): 101.668	Temperature (C): 26.63
Pressure (kPa): 101.664	Temperature (C): 26.63
Pressure (kPa): 101.662	Temperature (C): 26.63
Pressure (kPa): 101.660	Temperature (C): 26.62
Pressure (kPa): 101.665	Temperature (C): 26.61
Pressure (kPa): 101.665	Temperature (C): 26.61
Pressure (kPa): 101.668	Temperature (C): 26.61
Pressure (kPa): 101.664	Temperature (C): 26.60
Pressure (kPa): 101.659	Temperature (C): 26.60
Pressure (kPa): 101.667	Temperature (C): 26.61

Task 9A : The values change rapidly as I move my hand closer to the board and it is more noticeable from top down.

Proximity: 240	
Proximity: 238	
Proximity: 239	Proximity: 0
Proximity: 239	Proximity: 0
Proximity: 240	Proximity: 0
Proximity: 239	Proximity: 0
Proximity: 238	Proximity: 0
Proximity: 237	Proximity: 0
Proximity: 238	Proximity: 0
Proximity: 239	Proximity: 0
Proximity: 240	Proximity: 0
Proximity: 240	Proximity: 0
Proximity: 238	Proximity: 0
Proximity: 237	Proximity: 0
Proximity: 238	Proximity: 0
Proximity: 237	Proximity: 0
Proximity: 238	Proximity: 0

Task 9B: The right gesture was more easily recognised by the MCU.

```
Gesture test started  
RIGHT  
Gesture test started  
LEFT
```

Task 9C : Flashing the light directly to the board produced the most amount of change in the system.

73,44,39,138	4097,4097,4097,4097
79,48,43,151	4097,4097,4097,4097
75,44,40,140	4097,4097,4097,4097
76,46,41,144	4097,4097,4097,4097
78,47,42,148	4097,4097,4097,4097
73,44,39,139	4097,4097,4097,4097
79,47,43,151	4097,4097,4097,4097
75,44,40,141	4097,4097,4097,4097
76,46,41,144	4097,4097,4097,4097
78,47,42,148	4097,4097,4097,4097
73,44,39,139	4097,4097,4097,4097
79,48,43,151	4097,4097,4097,4097
74,44,39,140	4097,4097,4097,4097
76,46,41,144	4097,4097,4097,4097
78,47,42,148	4097,4097,4097,4097
75,45,41,143	4097,4097,4097,4097
79,47,43,151	4097,4097,4097,4097
73,43,39,138	4097,4097,4097,4097

Task 10: I initially started with very high Thresholds which made me unable to recreate most of the settings and then slowly started adjusting in accordance to the plots and serial readings that I had previously collected and then also based on the raw observations produced during the task.

```
raw,mic=17,clear=62,motion=0.010,prox=240
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=19,clear=65,motion=0.010,prox=239
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=6,clear=65,motion=0.010,prox=239
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=13,clear=61,motion=0.009,prox=242
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=6,clear=61,motion=0.008,prox=242
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
```

```
raw,mic=41,clear=1,motion=0.020,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=68,clear=1,motion=0.023,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=13,clear=1,motion=0.023,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=7,clear=1,motion=0.025,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
```

```
raw,mic=98,clear=55,motion=0.011,prox=239
flags,sound=1,dark=1,moving=0,near=1
state,NOISY_BRIGHT_STEADY_FAR
```

```
raw,mic=55,clear=108,motion=0.124,prox=183
flags,sound=1,dark=1,moving=1,near=0
state,NOISY_BRIGHT_MOVING_NEAR
```

Task 11: The thresholds were decided based on noticing nominal readings and then based on them and introduction of the different conditions noticing the changes in the system and then changing them to make it so that the conditions are met. As for the cooldown time it was selected by checking the previous tasks and seeing what was the minimal and optimal time that should work.

```
raw,rh=44.50,temp=25.49,mag=151.86,r=28,g=19,b=13,clear=52
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.32,temp=25.48,mag=151.54,r=28,g=20,b=13,clear=53
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.17,temp=25.48,mag=148.68,r=28,g=20,b=13,clear=53
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.09,temp=25.46,mag=153.43,r=32,g=22,b=15,clear=59
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.05,temp=25.48,mag=151.61,r=32,g=22,b=15,clear=59
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
```

```
raw,rh=67.80,temp=25.57,mag=150.02,r=33,g=22,b=15,clear=55
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
raw,rh=66.81,temp=25.52,mag=147.03,r=32,g=25,b=17,clear=61
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
raw,rh=64.74,temp=25.52,mag=150.71,r=32,g=25,b=17,clear=61
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
```

```
raw,rh=44.38,temp=26.40,mag=119.77,r=37,g=23,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.63,temp=26.40,mag=116.42,r=37,g=23,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.45,temp=26.40,mag=116.50,r=38,g=24,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.23,temp=26.40,mag=115.37,r=38,g=24,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
```

```
raw,rh=42.21,temp=26.26,mag=137.72,r=4097,g=4097,b=4097,clear=4097
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
raw,rh=42.28,temp=26.27,mag=138.18,r=4097,g=4097,b=4097,clear=4097
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
```